Assignment 4 – Normalization

Exercise 1: Parts Order Form

Assumptions:

1. A customer can place multiple orders but cannot place two different orders at the same date and the same time. Therefore, an order can be identified by customerNumber, orderDate, and orderTime together.
2. Each order is handled by one employee. An employee can help different customers, and a customer can be helped by different employees on different orders.
3. Each part only appears once in an order. If several units of the same part have been ordered, quantityOrdered will record the total quantity.

Original Data (1NF)

customerNumber	customerName	customerType	**orderDate**	**orderTime**	employee	**partNumber**	partName	partType	cageCode	quantityOrdered	unitPrice
HG54587	Jeff Peterson	Consumer	7/1/2024	10:30 am	D. Harrison	10654	Float Control	Plumbing	G413	4	12
HG54587	Jeff Peterson	Consumer	7/1/2024	10:30 am	D. Harrison	10456	Modulator	Electrical	H433	3	7
HG54587	Jeff Peterson	Consumer	7/1/2024	10:30 am	D. Harrison	10776	Hose Assembly	Plumbing	G413	7	9
HG54587	Jeff Peterson	Consumer	7/1/2024	10:30 am	D. Harrison	10657	Float Assembly	Plumbing	G413	5	10

This table is in 1NF because each cell contains a single value and each row represents one part within an order. The customer and order information is repeated for each ordered part. The primary key of this table is (customerNumber, orderDate, orderTime, partNumber). The first three identify an order, and partNumber identifies a particular part within that order.

For the normalization process, I assume that the customerNumber determines customerName and customerType, and the partNumber determines partName, partType, and a fixed unitPrice. Prices do not vary between orders in this model. I also assume that each partType is assigned to one cageCode and all parts of that type are stored in that cage.

This table is not in 2NF because some non-key attributes only depend on part of the composite primary key. The customerName and customerType only depend on customerNumber. The employee depends on the order, identified by customerNumber, orderDate, and orderTime. The partName, partType, cageCode, and unitPrice depend only on partNumber. However, the quantityOrdered depends on the complete key because it records the quantity of a specific part in a specific order. I divide the original table into four relations (customer, order_header, part, and order_line) to remove these partial dependencies.

Second Normal Form (2NF)

customer:

customerNumber	customerName	customerType
HG54587	Jeff Peterson	Consumer

order_header:

customerNumber	**orderDate**	**orderTime**	employee
HG54587	7/1/2024	10:30 am	D. Harrison

part:

partNumber	partName	partType	cageCode	unitPrice
10654	Float Control	Plumbing	G413	12
10456	Modulator	Electrical	H433	7
10776	Hose Assembly	Plumbing	G413	9
10657	Float Assembly	Plumbing	G413	10

order_line:

customerNumber	**orderDate**	**orderTime**	**partNumber**	quantityOrdered
HG54587	7/1/2024	10:30 am	10654	4
HG54587	7/1/2024	10:30 am	10456	3
HG54587	7/1/2024	10:30 am	10776	7
HG54587	7/1/2024	10:30 am	10657	5

These tables have removed the partial dependencies in the original table. In the customer table, customerNumber determines the customerName and the customerType. In the order_header table, the employee depends on the complete order key: customerNumber, orderDate, and orderTime. In the part table, the partNumber determines partName, partType, cageCode, and unitPrice. In the order_line table, the quantityOrdered depends on all four fields of the composite primary key since the quantity is specific to both an order and a part. Each non-key attribute now depends on the whole primary key of its relation, so all 4 relations are in 2NF.

Third Normal Form (3NF)

According to the storage assumption, partNumber→partType→cageCode is a transitive dependency in the part table. I move the relationship between partType and cageCode into an individual part_type table, and remove cageCode from part to reach 3NF. The partType will remain in part as a foreign key, and the other 3 relations retain the same attributes.

customer:

customerNumber	customerName	customerType
HG54587	Jeff Peterson	Consumer

order_header:

customerNumber	**orderDate**	**orderTime**	employee
HG54587	7/1/2024	10:30 am	D. Harrison

part:

partNumber	partName	*partType*	unitPrice
10654	Float Control	Plumbing	12
10456	Modulator	Electrical	7
10776	Hose Assembly	Plumbing	9
10657	Float Assembly	Plumbing	10

part_type:

partType	cageCode
Plumbing	G413
Electrical	H433

order_line:

customerNumber	***orderDate***	***orderTime***	***partNumber***	quantityOrdered
HG54587	7/1/2024	10:30 am	10654	4
HG54587	7/1/2024	10:30 am	10456	3
HG54587	7/1/2024	10:30 am	10776	7
HG54587	7/1/2024	10:30 am	10657	5

In the order_header table, customerNumber is a foreign key which references the customerNumber in customer. In the part table, partType is a foreign key which references the partType in part_type. The order_line table has 2 foreign keys: the first is (customerNumber, orderDate, and orderTime), which references the matching 3-field primary key in order_header. The second one is partNumber, which references partNumber in part. The five relations have no partial or transitive dependencies of non-key attributes on their keys, so they are in 3NF.

Exercise 2: Therapist Appointments
Assumptions:

1. The "staffNo" only identifies a single unique therapist, and the "patNo" only identifies a single unique patient, but the name may not be unique.
2. Each row represents one appointment between a therapist and a patient. I assume that the same therapist cannot have two different appointments on the same date and at the same time. Therefore, (staffNo, appointmentDate, appointmentTime) together can uniquely identify one appointment.
3. The same therapist only sees patients at one branch on the same day, so (staffNo, appointmentDate) can identify the branch

number. This therapist can work at different branches on other dates.

4. A patient can have multiple appointments on the same day, and can also make appointments with different therapists. The fields "appointmentDate" and "appointmentTime" represent the existing dates and times in the original table, and no new attributes have been added.

Original Data (1NF)

staffNo	therapistName	patNo	patName	**appointmentDate**	**appointmentTime**	branchNo
S1011	Fred Smith	P100	Lily White	9/12/2022	10:00	M15
S1011	Fred Smith	P105	Jill Baker	9/12/2022	12:00	M15
S1024	Heidi Pierce	P108	Andy McKee	9/12/2022	10:00	Q10
S1024	Heidi Pierce	P108	Andy McKee	9/14/2022	14:00	Q10
S1032	Richard Levin	P105	Jill Baker	9/14/2022	16:30	M15
S1032	Richard Levin	P110	Jimmy Winter	9/15/2022	18:00	B13

Each cell in the original table has only one value, thus it complies with 1NF. Its composite primary key is (staffNo, appointmentDate, appointmentTime). The main functional dependencies are: staffNo → therapistName; patNo → patName; (staffNo, appointmentDate) → branchNo; (staffNo, appointmentDate, appointmentTime) → patNo.

This table does not meet 2NF either, because therapistName only depends on staffNo, and branchNo only depends on (staffNo, appointmentDate). Both of them only depend on a part of the composite primary key, which belongs to partial dependency.

Second Normal Form (2NF)

I divided the original table into three relations to eliminate partial dependency: therapist stores the name of the therapist; therapist_schedule stores the branch where each therapist works on each date; appointment stores the date, time and patient information of each appointment.

therapist:

staffNo	therapistName
S1011	Fred Smith
S1024	Heidi Pierce
S1032	Richard Levin

therapist_schedule:

staffNo	**appointmentDate**	branchNo
S1011	9/12/2022	M15
S1024	9/12/2022	Q10
S1024	9/14/2022	Q10
S1032	9/14/2022	M15
S1032	9/15/2022	B13

appointment:

staffNo	**appointmentDate**	**appointmentTime**	patNo	patName
S1011	9/12/2022	10:00	P100	Lily White
S1011	9/12/2022	12:00	P105	Jill Baker
S1024	9/12/2022	10:00	P108	Andy McKee
S1024	9/14/2022	14:00	P108	Andy McKee
S1032	9/14/2022	16:30	P105	Jill Baker
S1032	9/15/2022	18:00	P110	Jimmy Winter

In the therapist table, therapistName depends on the complete Primary Key staffNo. In the therapist_schedule table, branchNo depends on the combination of staffNo and appointmentDate. In the appointment table, both patNo and patName depend on the complete Primary Key, and there is no partial dependency anymore. Therefore, these three relations all conform to 2NF. However, the appointment still has a transitive dependency: (staffNo, appointmentDate, appointmentTime) → patNo → patName. This means

that the name of the same patient will still repeat in multiple appointments.

Third Normal Form (3NF)

To eliminate the transitive dependency, I created the "patient" relation, using "patNo" as the Primary Key, and moved "patName" into this relation. The "appointment" relation retains "patNo" as the Foreign Key. The attributes of "therapist" and "therapist_schedule" remain unchanged.

therapist:

staffNo	therapistName
S1011	Fred Smith
S1024	Heidi Pierce
S1032	Richard Levin

patient:

patNo	patName
P100	Lily White
P105	Jill Baker
P108	Andy McKee
P110	Jimmy Winter

therapist_schedule:

staffNo	**appointmentDate**	branchNo
S1011	9/12/2022	M15
S1024	9/12/2022	Q10
S1024	9/14/2022	Q10
S1032	9/14/2022	M15
S1032	9/15/2022	B13

appointment:

staffNo	***appointmentDate***	**appointmentTime**	*patNo*
S1011	9/12/2022	10:00	P100
S1011	9/12/2022	12:00	P105
S1024	9/12/2022	10:00	P108
S1024	9/14/2022	14:00	P108
S1032	9/14/2022	16:30	P105
S1032	9/15/2022	18:00	P110

The staffNo in the therapist_schedule table references the staffNo of the therapist. In the appointment table, the staffNo and appointmentDate together form a composite Foreign Key that references the composite Primary Key (staffNo, appointmentDate) of the therapist_schedule table; patNo is another Foreign Key that references the patNo of the patient. appointmentTime is part of the composite primary key of the appointment table, but not of any of the above Foreign Keys.
All these relations comply with 3NF because each non-key attribute depends on the complete Primary Key of the respective relation, and there are no transitive dependencies.

Exercise 3: Event Staff

Assumptions:

1. I assume that each eNo corresponds to an eName, but different employees can have the same name, so I do not use eName as the Primary Key.
2. I assume that contractNo uniquely identifies a contract. Each contract corresponds to only one event, while the same event can have multiple contracts. Therefore, contractNo → eventNo.
3. I assume that eventNo identifies a unique event, and each event has only one eventLoc. Therefore, eventNo → eventLoc. Different events can occur at the same location, so eventLoc is not necessarily unique.
4. I assume that hours represent the cumulative working hours of an employee under a contract. Therefore, each (eNo, contractNo) combination is unique. Thus, (eNo, contractNo) is the composite primary key of the original table.

Original Data (1NF)

eNo	**contractNo**	hours	eName	eventNo	eventLoc
1135	C1024	16	Smith J	H25	Queens
1057	C1024	24	Hocine D	H25	Queens
1068	C1025	28	White T	H4	Yonkers
1135	C1025	15	Smith J	H4	Yonkers
1135	C1026	10	Smith J	H25	Queens

Each cell in the original table contains only one value, and each row represents the work record of an employee under a contract. Therefore, it complies with 1NF. The eNo alone cannot identify a single record because the same employee can participate in multiple contracts; the contractNo alone cannot identify a single record either because the same contract can involve multiple employees. Together, they form the Primary Key.

The main functional dependencies of this table are: eNo → eName; contractNo → eventNo; eventNo → eventLoc; (eNo, contractNo) → hours.

This table does not conform to 2NF because eName only depends on eNo, and eventNo and eventLoc only depend on

contractNo . This means that they all belong to partial dependencies. Hours, however, depends on the complete composite Primary Key.

Second Normal Form (2NF)

I divided the original table into three relations to eliminate partial dependency: "employee" stores the employee's name; "contract" stores the event and location corresponding to each contract; "employee_contract" stores the hours of each employee for each contract.

employee:

eNo	eName
1135	Smith J
1057	Hocine D
1068	White T

contract:

contractNo	eventNo	eventLoc
C1024	H25	Queens
C1025	H4	Yonkers
C1026	H25	Queens

employee_contract:

eNo	**contractNo**	hours
1135	C1024	16
1057	C1024	24
1068	C1025	28
1135	C1025	15
1135	C1026	10

In the "employee" table, "eName" depends on the complete Primary Key "eNo". In the "contract" table, "eventNo" and "eventLoc" depend on the complete Primary Key "contractNo". In the "employee_contract" table, "hours" depends on the combination of "eNo" and "contractNo", and cannot be determined by either attribute alone. Therefore, these three relations no longer have partial dependency and all conform to 2NF. However, there is still a transitive dependency in the contract: contractNo → eventNo → eventLoc. eventLoc describes the event, not the contract itself. When the same event corresponds to multiple contracts, the location will be duplicated and saved, so the contract relation is not yet in 3NF.

Third Normal Form (3NF)

I created an "event" table with "eventNo" as the Primary Key, and moved "eventLoc" into this relation. In the "contract" table, I removed "eventLoc" and retained "eventNo" as the Foreign Key. I kept the attributes of the "employee" and "employee_contract" tables unchanged.

employee:

eNo	eName
1135	Smith J
1057	Hocine D
1068	White T

event:

eventNo	eventLoc
H25	Queens
H4	Yonkers

contract:

contractNo	*eventNo*
C1024	H25
C1025	H4
C1026	H25

employee_contract:

eNo	***contractNo***	hours
1135	C1024	16
1057	C1024	24
1068	C1025	28
1135	C1025	15
1135	C1026	10

In the "contract" table, the "eventNo" column references the "eventNo" of the "event" table. In the "employee_contract" table, there are two independent Foreign Keys: "eNo" references the "eNo" of the "employee" table, and "contractNo" references the "contractNo" of the "contract" table. "eNo" and "contractNo" together form the composite Primary Key of "employee_contract", but they are not a composite Foreign Key. All these four relations comply with 3NF: each non-key attribute depends on the complete Primary Key of the respective relation, and there are no transitive dependencies.